



XRPL Make Waves Hackathon

NFTs You Can Rebuild, Not Just Collect

Mint characters, wear and trade their traits, and buy and sell on the XRP Ledger — every step a native XRPL transaction.

LFG is a live, non-custodial, open-source NFT platform on XRPL mainnet since July 10, 2026 — five surfaces across the web, Discord, and Telegram.



Try the live app

build.letseffinggo.com

Team: LFG · Joshua Hamsa · hamsa@letseffinggo.com



What will YOU build?

The Closet: a real character beside the trait inventory it can be rebuilt from.



The XRPL Has the Primitives. It Doesn't Have the Examples.

Two gaps

1 • Builders have tools, not examples.

XRPL ships strong primitives — NFTTokenModify, native NFTTokenOffers, TokenEscrow, the DEX and AMM — but almost nothing open source shows them assembled into a real product. Nothing to fork, nothing to point an agent at. Adoption is limited by examples, not tools.

2 • Collectors have to leave their community.

Interacting with NFTs on XRPL has meant minting on a launchpad, trading on someone's marketplace, and nothing at all where the community actually lives — Discord, Telegram, a social feed. The NFT itself is a finished image: ownership persists, but participation fades.

“The tools were ready. The example was missing.”

Shipped on the XRP Ledger



NFTTokenModify



NFTTokenOffer



TokenEscrow



DEX + AMM



Memos + SourceTag



Clio history

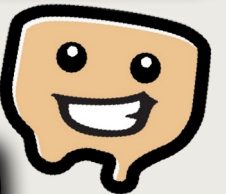
Open-source products that assemble them



No end-to-end reference to fork. No codebase to hand an agent. Every team re-derives the same flows from the docs.

Where collectors are today: mint on a launchpad → trade on a marketplace → community chat has no part in it.

Nothing to fork?



LFG Is the Example: a Complete NFT Platform From Primitives Alone

With LFG, collectors can:

- **Mint** dynamically composed characters — singly or in bulk, paying LFGO or plain XRP.
- **Swap traits** between NFTs they own — in place, on-ledger.
- **Harvest** a character into an on-chain Closet (a soulbound inventory NFT) and **Build** new looks from it.
- **Sell traits straight from the Closet** — no signature to list — and **bid** on traits they want.
- **Buy and sell characters** for XRP, with bids and fee cover.
- **Earn BRIX daily** for holding, and claim it on-chain.
- **Use their own wallet** — Xaman or Joey — from the web, Discord, or Telegram.

LFG creates a continuing cycle of collecting, building, and trading after the initial mint.

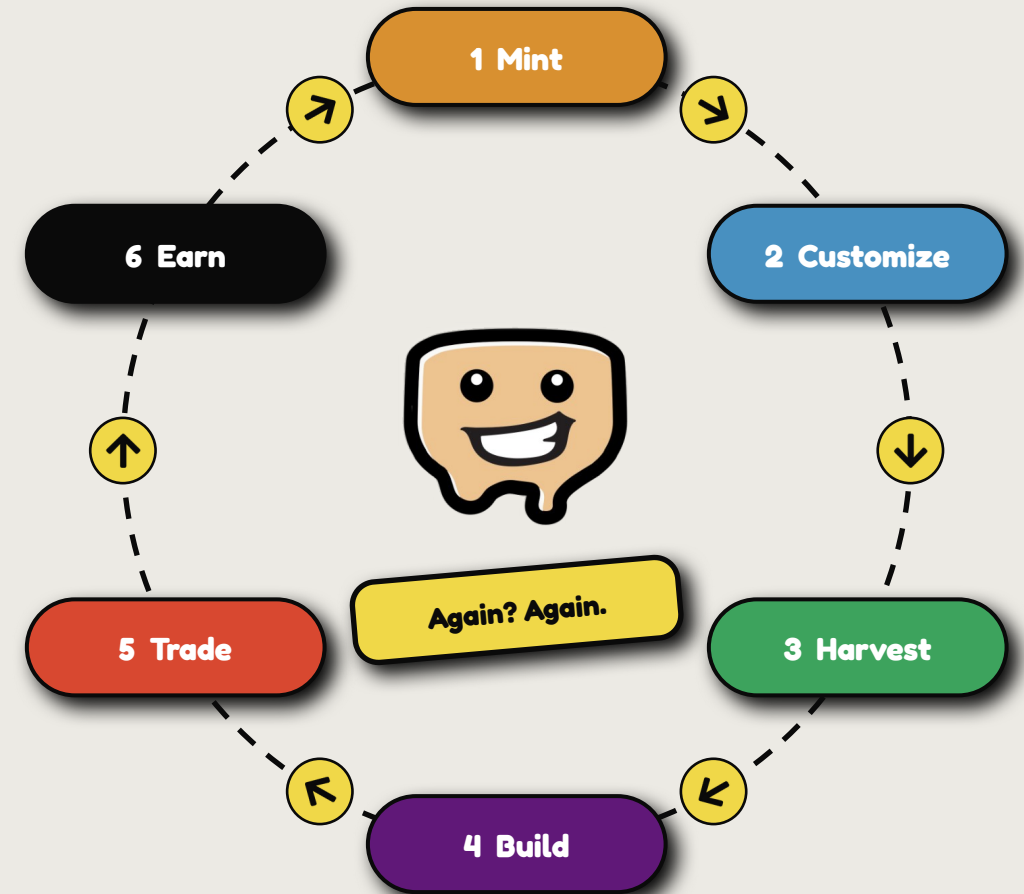


Restyle: a live character (#59, skeleton body) beside the owner's Closet. Tap a trait, save once, and NFTTokenModify updates the same token in place — no burn, no re-mint.

Every Action Creates a Reason to Return

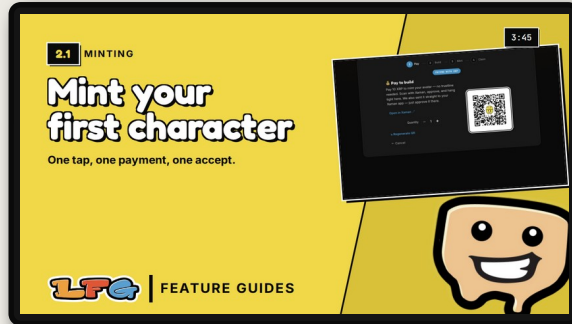
- 1 Mint** Generate a layered character. Holders pay LFGO, which is burned; XRP mints trigger a best-effort LFGO buy-and-burn on the DEX.
- 2 Customize** Swap traits on-ledger with NFTTokenModify.
- 3 Harvest** Move a character's traits into your Closet.
- 4 Build** Dress a blank or existing character from the Closet — one NFTTokenModify per save.
- 5 Trade** Characters for XRP; traits for BRIX on the Closet Market (asks, bids, fills).
- 6 Earn** 1 BRIX per unlisted NFT per day, claimed on-chain and spent back into traits.

“Instead of ending at mint, the experience begins there.”



One Complete NFT Platform, Live Across Five Client Surfaces

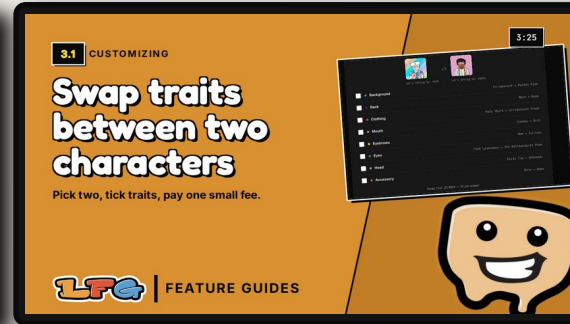
Creation



One tap, one payment, one accept — the art composes from trait layers and reveals in the app.

- Dynamic + animated generation
- Single, bulk, and sponsored free mints
- Rarity rules with a public mint-odds view

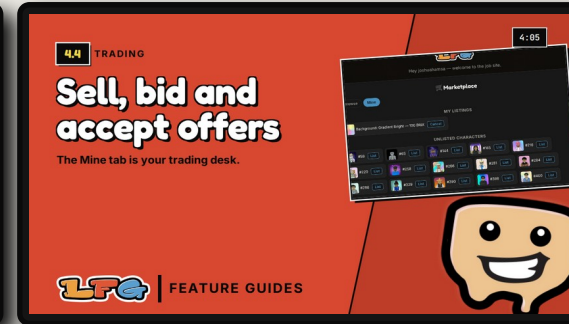
Customization



Pick two characters, tick traits, pay one small fee; both NFTs update in place.

- In-place trait swapping
- Closet, Harvest, Build, Equip
- Animated trait layers

Commerce



The Mine tab is the trading desk: list, bid, and accept offers on characters and traits.

- Character marketplace with bids + fee cover
- Closet Market: asks and escrow-backed bids
- XRP→BRIX AMM on-ramp · xrp.cafe Buy now

Community



Who holds, swaps, builds, and earns the most — eight boards from the ledger history.

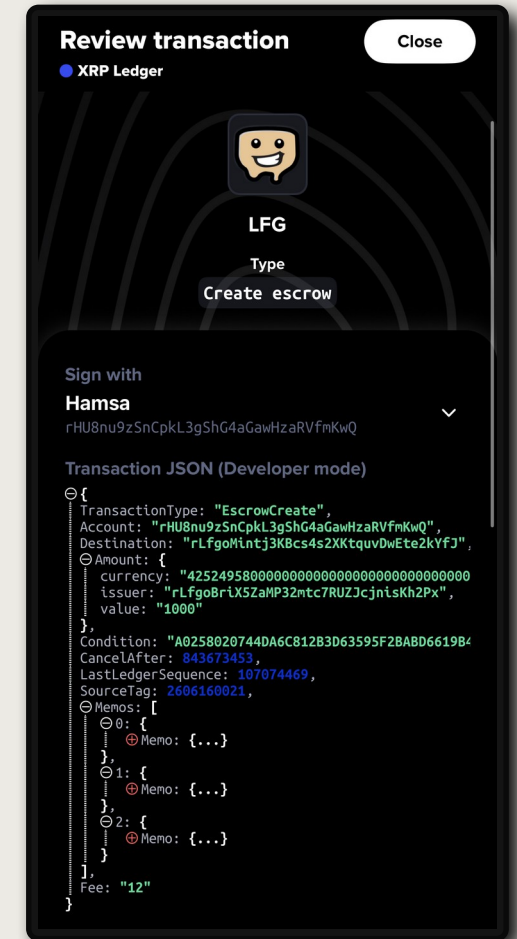
- BRIX daily drip for holders
- Share cards open the app in any in-app browser (X, LinkedIn, Reddit...) + X mint auto-poster
- Discord, Telegram, and an installable web app

Thumbnails are from the LFG feature-guide series (youtube.com — same flows, recorded on the live app).

LFG Uses the Ledger as Infrastructure – With No Smart Contracts

XRPL primitive	How LFG uses it
NFTTokenModify	In-place trait swaps, Harvest, Build and Equip (XLS-46) — and the soulbound Closet whose on-chain metadata is the inventory of record
NFTTokenOffer	Non-custodial character and trait trading: native sell offers and bids, no escrow contract, no marketplace custody
TokenEscrow	Trait bids lock BRIX under a PREIMAGE-SHA-256 condition only the fill can release (XLS-85) — a collection offer for any trait
Issued currencies	LFGO: the mint price, burned; XRP mints buy-and-burn it best-effort off the DEX. BRIX prices traits, pays swap fees, and drips daily to holders
DEX + AMM paths	XRP payers join with no trust line — the backend buys LFGO off the DEX order book (best-effort) and clears XRP into BRIX through the BRIX/XRP AMM
SourceTag + Memos	Every transaction records the app, the signer, the surface, and the action
Clio history	Ownership, activity, leaderboards, drip accrual, and free-mint eligibility all derive from a certified ledger archive

“Every core interaction is a verifiable XRPL transaction – built from primitives the ledger already ships.”



*A real trait bid as reviewed in Xaman:
EscrowCreate for 1,000 BRIX with a crypto-
condition, SourceTag 2606160021, and
provenance memos.*

Traits Live in Your Closet and Trade Without Leaving the Ledger



1

Harvest

The character is stripped to a blank in place (NFTokenModify); its traits are credited to your Closet — a soulbound mutable NFT whose metadata is the source of truth.



2

Build / Equip

Dress a blank or swap one slot from the Closet; one NFTokenModify per save.



3

Sell or Bid

List a trait from the Closet with zero signatures — the ask encumbers it and is mirrored into the Closet NFT on the next sync. Bid on any trait by locking BRX in a TokenEscrow.

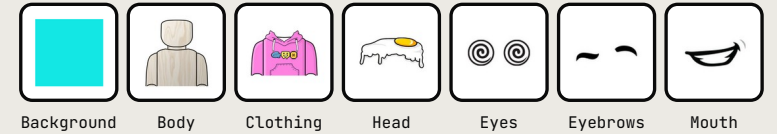


4

Fill

A Payment or EscrowFinish settles it; the trait moves Closet → Closet in one step and both Closet NFTs are re-synced on-ledger.

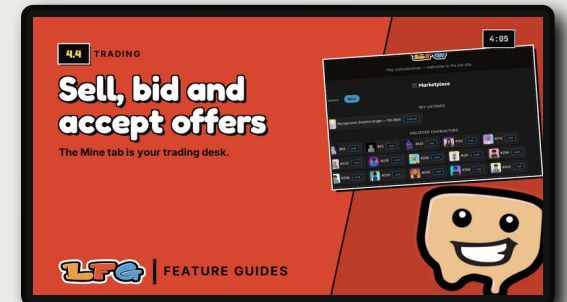
- A conservation ledger and nightly reconcile-and-audit guard trait supply; compatibility rules prevent invalid combinations across five body types.
- Trait tokens extracted under the earlier model still work — Deposit returns them to a Closet.



Closet — seven separately owned trait assets, credited by Harvest (real layer art from the LFG API).



Your Closet: take characters apart into reusable parts, for free.



Sell, bid, and accept offers — asks are free, bids lock BRX in escrow.

Listed. That's it.

Snappy for the user, fully accounted for on the ledger.

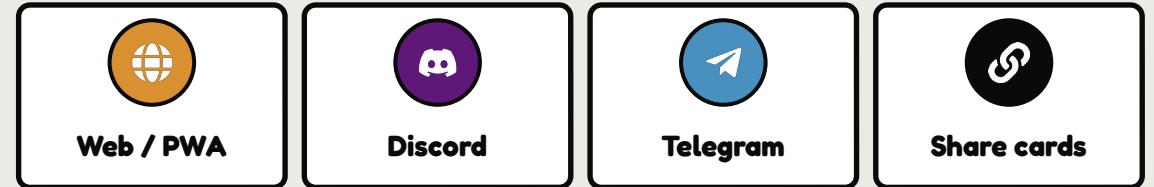


Users Keep Control of Their Wallets and Assets

Key protections

-  Private keys never enter LFG.
-  Users sign in their own wallet — Xaman, or Joey on the web; the backend signs only project operations.
-  Signing requests are pinned to the initiating wallet.
-  Native offers and TokenEscrows — user assets never sit in an LFG wallet; Closet Market fills settle by the backend releasing the on-ledger escrow.
-  Durable, resumable sessions; a simulate pre-flight before any backend transaction.
-  Listeners verify outcomes; unknowns resolve only by memo lookup or ledger expiry.
-  Provenance memos identify every action and originating surface.

One shared backend provides consistent security and behavior across every client.

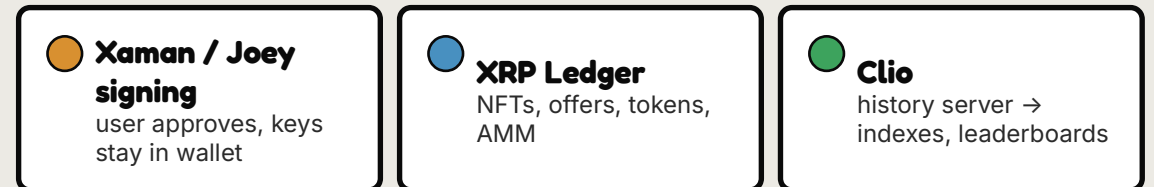


Users · sign in with their wallet · one identity across surfaces



Shared LFG service

REST/WS hub · session state machines · builds every signing payload · submits every transaction



Simplified view. Every surface is a thin client; the shared service holds no private keys and every ledger write is signed by the user in Xaman. The full system diagram is in Appendix A.

Xaman — the wallet used to approve LFG transactions (formerly XUMM); **Joey** — a second wallet supported on the web via WalletConnect. **Clio** — the XRPL history server LFG reads ownership and activity from.

LFG Is a Working Mainnet Product With Verifiable Usage

5,200+

Live NFT editions

Cut over at 3,535 on July 10, 2026;
edition #5,247 minted Sep 20

12,128

SourceTag-attributed XRPL transactions

Mints, offers, accepts, modifies,
payments, escrows, trust lines

718

Unique participating wallets

405 unique actors after funder de-
duplication; operator wallets excluded

5

Client surfaces

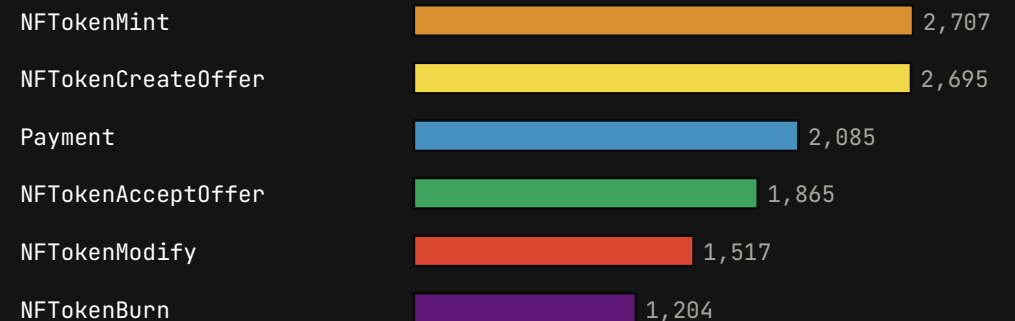
Discord bot + Activity, Telegram bot +
Mini App, web app

Ledger metrics captured Sep 21, 2026 · Source: XRPL SourceTag 2606160021 (metrics/sourcetag.json in the repo)

Additional proof

- Mainnet collection July 10 · trait economy July 21 · X auto-poster Sep 10 · Closet Market with TokenEscrow bids Sep 18, 2026.
- Character marketplace, Closet Market, and BRIX daily drip are live.
- Every LFG transaction uses SourceTag **2606160021**.
- 5,376 automated tests · 333 pull requests merged · 774 commits during the sprint.

Tagged transactions by type

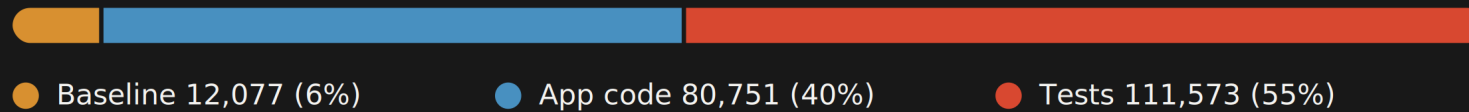


+55 TrustSet / EscrowCreate / EscrowFinish / CancelOffer / AccountSet · source:
metrics/sourcetag.json, Sep 21, 2026

The Hackathon Turned a Mint Bot Into a Multichannel On-Chain Economy

Code written during the hackathon

since June 21 · +192,324 net lines · codebase now 204,401 lines



12,077

application lines at the pre-hackathon baseline

80,751

application lines added during the hackathon

111,573

test lines added during the hackathon

Repository-generated code-growth graphic, regenerated on every push to main. Baseline measured at commit e296308 (June 19, 2026); hackathon began June 21.

333 PRs · 774 commits · source:
assets/hackathon_loc.svg + dashboard.svg, Sep 21, 2026

During the hackathon, LFG added or substantially expanded:



Shared service architecture + five surfaces



Trait economy: Closet, Harvest, Build, Equip



Closet Market: zero-signature asks + TokenEscrow bids



Marketplace with bids, fee cover, external Buy now



LFGO mint burn + DEX buyback · AMM on-ramp · BRIX drip



Bulk minting · animated NFTs · sponsored free mints



Xaman push signing + Joey Wallet



Ledger indexer · 8 leaderboards · SourceTag metrics



Share-card funnel (any in-app browser) + X auto-poster



Mainnet deploy · staging/prod promotion · nightly audits

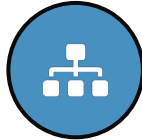
“Not a prototype – a deployed product with users and ledger activity.”

Fork It, or Point an Agent at It: LFG Is an MIT Reference Stack



MIT licensed

The whole stack — mint engine, trait rules, Closet model, marketplace and Closet Market flows, indexer, drip — is open source under MIT. No smart contracts to audit; every flow is a native transaction you can read.



One backend, thin surfaces

A shared-services spine (lfg_service) plus a Surface SDK: Discord, Telegram, and the web are thin clients. Adding a surface is writing a client, not re-implementing the ledger.



Fork it for your collection

Point trait_config.yaml at your art, set your issuer and tokens, and the same mint / swap / Closet / market flows run for your community.



Or point an agent at it

CLAUDE.md, design specs, implementation plans, and a merged-PR build log document every decision — a codebase an agent can read, extend, and test against 5,000+ tests.

Make Waves opportunities: OPP-027 Dynamic NFT Framework, OPP-053 transaction indexer, OPP-046 gaming utility NFTs, OPP-001 session continuity, OPP-062 community ops — partial (collection-scoped; rationale in docs/HACKATHON.md). OPP-016 POAP and OPP-074/019 dev utilities — buildable from what ships.

“The goal is the toolkit: what the XRPL needs to create engaging, fun experiences for participants.”

Mint It. Rebuild It. Trade It. Fork It.



Try the live app



build.letseffinggo.com



Review the source



github.com/Team-Hamsa/LFG



Watch the demo



youtu.be/bZ12oJGYfN0



Fork it



github.com/Team-Hamsa/LFG/fork

“The tools were ready. Now there’s an example.”

LFG · Joshua Hamsa · hamsa@letseffinggo.com · letseffinggo.com · MIT license · SourceTag 2606160021



Detailed Architecture: Five Surfaces, One Service, Ledger Listeners

- 1 User-facing surfaces**
 Discord bot, Discord Activity, Telegram bot and Mini App, and the web app — thin clients over REST/WebSocket.
- 2 Shared LFG service**
 aiohttp hub: auth and identity, session state machines, every Xaman payload, every ledger submission.
- 3 Domain flow modules**
 Mint, bulk mint, swap, economy, market, Closet Market, and shop flows in one shared library with the trait engine and rarity rules.
- 4 Clio listeners and databases**
 A separate process group streams the Clio transaction feed into per-network SQLite index and history stores.
- 5 External systems**
 XRP Ledger (Clio and rippled), Xaman and Joey signing, BunnyCDN image and metadata hosting.

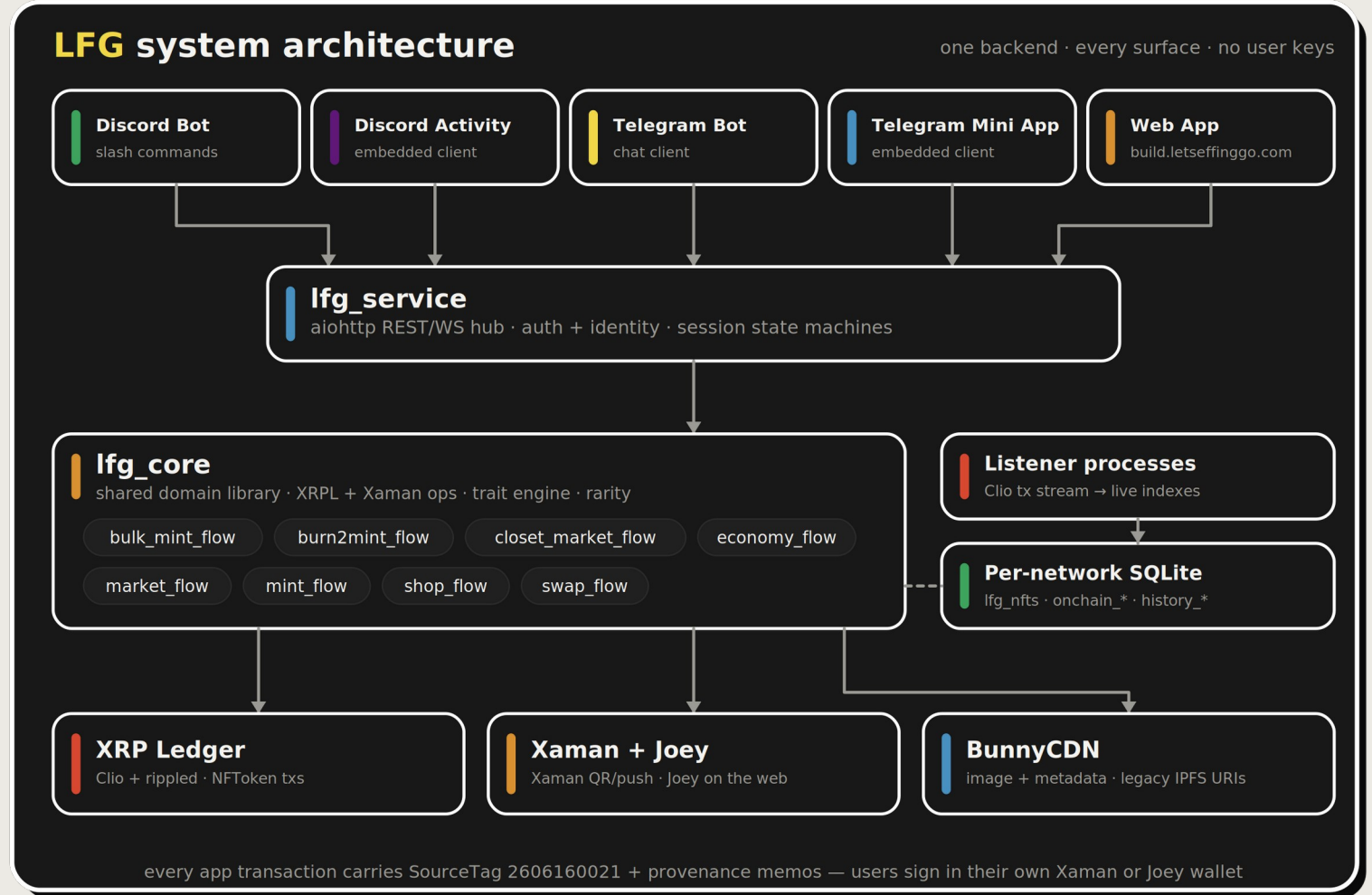
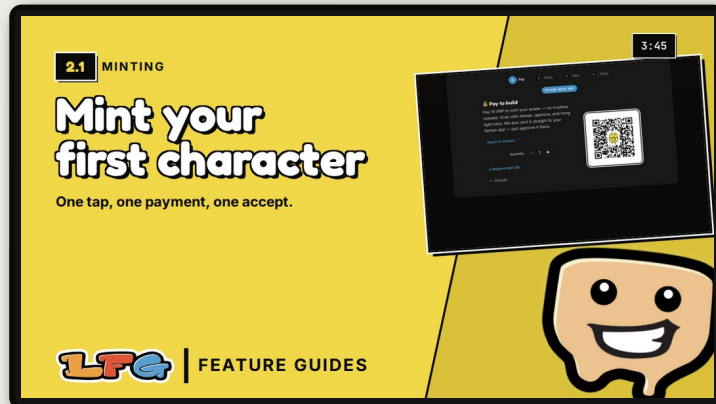


Diagram: assets/architecture.svg in the repository (github.com/Team-Hamsa/LFG). Every XRPL transaction carries SourceTag 2606160021 plus provenance memos; all signing happens in the user's Xaman wallet.

Product Gallery: The Core Flows



Mint — pick a surface, pay in LFGO or XRP, sign once; the art composes from trait layers and reveals in the app.



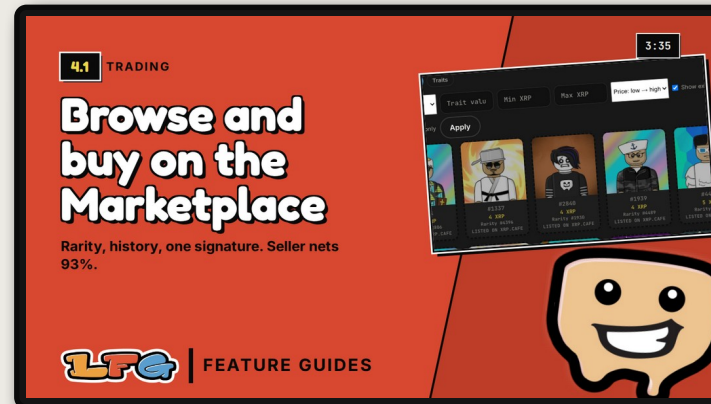
Trait Swapper — two characters you own trade a trait; NFTTokenModify updates both in place, same token IDs.



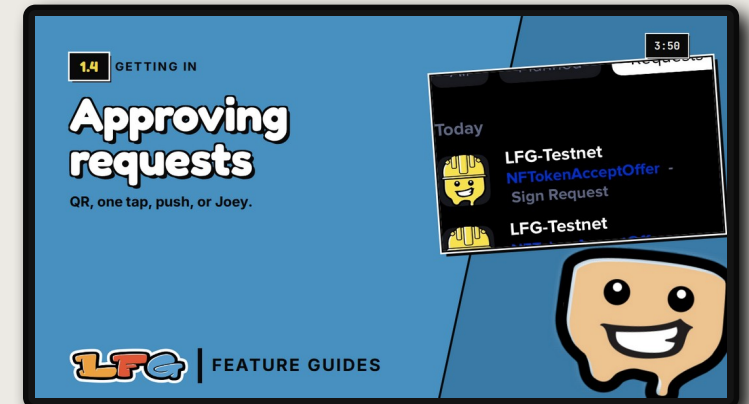
Restyle — preview Closet parts on a live character, save once; one NFTTokenModify per save.



Buy traits — Closet asks and trait listings priced in BRIX; no BRIX? one extra tap pays XRP via the AMM.



Marketplace — browse by rarity and history, buy in one signature on a native NFTTokenOffer; seller nets 93%.



Approving requests — every LFG action arrives in your own wallet as a sign request, pushed or by QR (shown on testnet).

Frames from the LFG feature-guide series. Demo video: youtu.be/bZ12oJGYfN0

Economic Flow: Three Currencies, Each With One Job

LFGO

Minting

- The mint price; burned on arrival.
- Single mints paid in XRP: the backend makes a best-effort buy of 1 LFGO off the DEX (there is no XRP/LFGO AMM) with at most the XRP just paid, and burns it only if that fills — no trust line needed to join.

BRIX

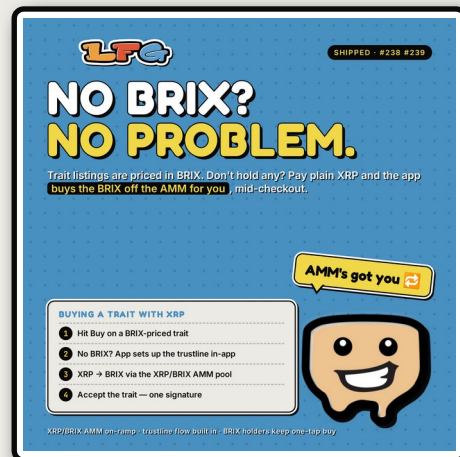
Trait economy

- Prices trait asks and bids; pays swap fees.
- Bids lock BRIX in a TokenEscrow until filled or cancelled.
- Drips to holders: 1 per unlisted NFT per day, claimed on-chain.

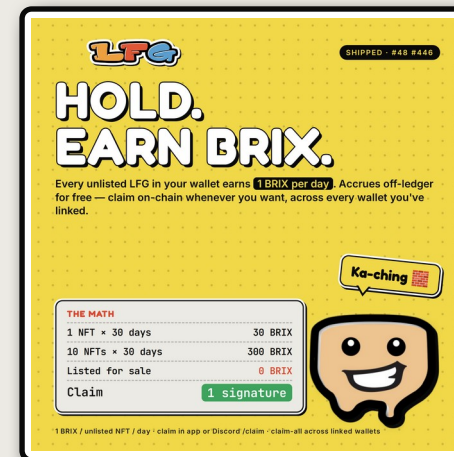
XRP

Characters + on-ramp

- Character trades on native NTokenOffers.
- On-ramp into BRIX through the BRIX/XRP AMM for buyers who hold none.



No BRIX? The app quotes the BRIX/XRP AMM and the buyer pays XRP; the payment path clears into BRIX mid-checkout.



Hold, earn: every unlisted NFT accrues 1 BRIX per UTC day from the ledger archive; claiming is one signature from the distributor wallet.

Mechanics only — nothing on this slide is a price prediction or investment claim.

Reliability and Testing: Built to Hold Real Value on Mainnet

Built in a hackathon sprint

live repo vitals · auto-updated on every push to main

5,376

tests

228

modules

774

commits

5

surfaces

commits / day since June 21



Repository vitals graphic (assets/dashboard.svg), regenerated on every push to main. Captured Sep 21, 2026.



Certified ledger archive

Per-network databases with a continuity-certified history archive and self-healing catch-up; a startup network guard separates mainnet from testnet.



Staging-to-production promotion

Merges deploy to staging automatically; production is an explicit, selective promotion step with endpoint failover in front of every ledger call.



Continuous integration checks

Lint, type checks, secret scanning, trait-config validation, and layout checks run on every push.



5,376 automated tests

The full suite runs locally at pre-push and again in CI; both block on failure.



Replay-safe payment handling

Payments are watched over the XRPL websocket and claimed once, so a restart can never double-charge or double-mint.



Durable session recovery

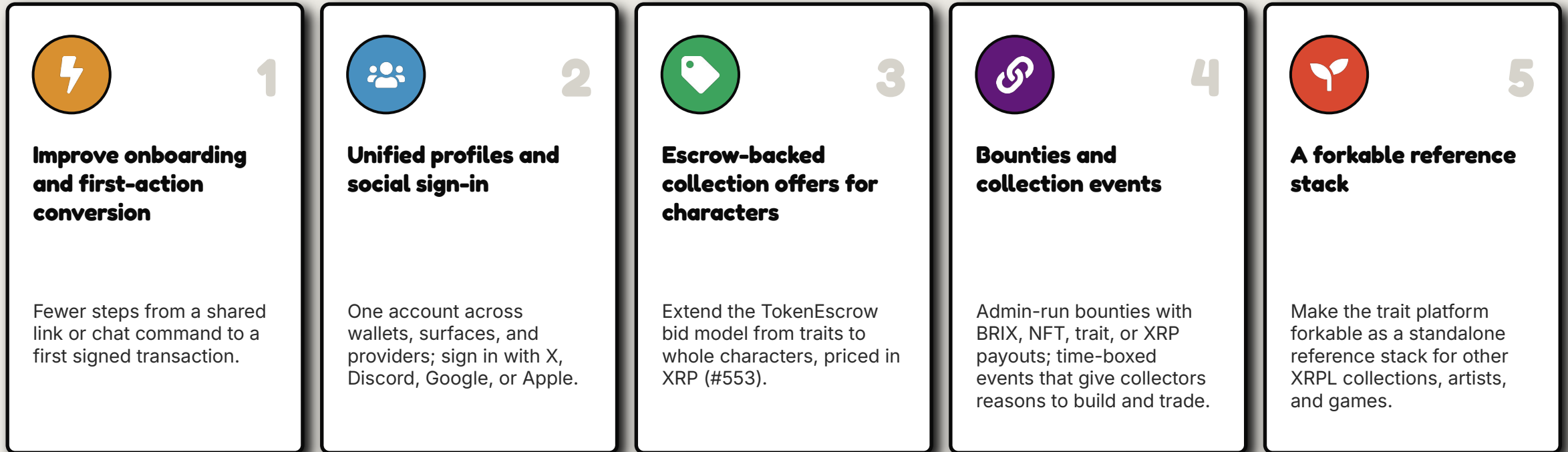
Mint, swap, market, economy, and Closet Market fills are journaled and resume after a restart.



Supply-conservation audits

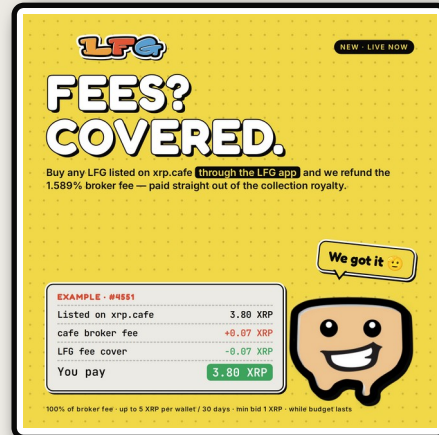
A conservation ledger plus a nightly reconcile-then-audit job guard trait supply integrity.

Roadmap: Grow Participation, Then Make the Stack Forkable



Outcomes, not engineering tasks. Open work is tracked publicly as roadmap-labelled issues — including XLS-56 Batch for single-signature multi-offer accepts (#219).

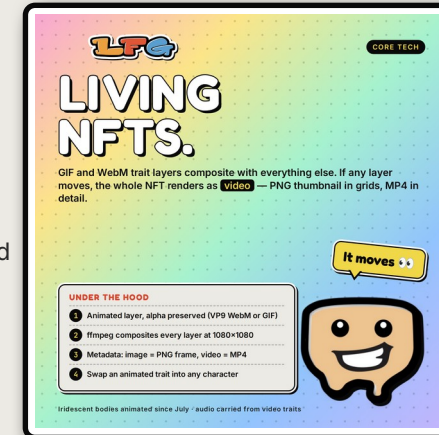
More That Shipped: Distribution, Identity, and Market Plumbing



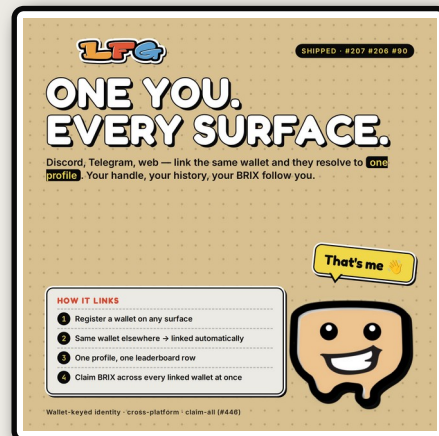
Fee cover — buy any LFG listed on xrp.cafe through the app and the 1.589% broker fee is refunded from the collection royalty.



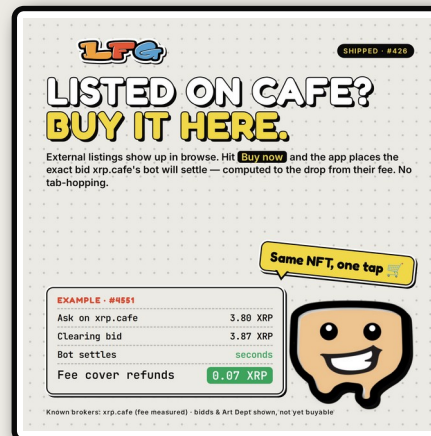
Telegram — register, mint, and swap from an inline-keyboard chat, or open the full Mini App; same backend as Discord and web.



Living NFTs — GIF/WebM trait layers composite with everything else; an animated layer makes the whole NFT render as video.



One profile — link the same wallet on Discord, Telegram, and web; one handle, one history, claim BRIX across every linked wallet.



Buy external listings — xrp.cafe listings show in Browse with a clearing-price Buy now computed to the drop from the broker fee.



Bids on anything — native on-ledger buy offers on any LFG, listed or not; owners accept from a pending-offers tray.

Not shown: Trait Shop (built, off in production — the project never sells traits) and burn-to-mint (built, flag-gated).